Person Detection with YOLOv8s

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This report covers a fine-tuned YOLOv8s model that finds people and draws a box around each one. Training used the COCO 2017 images that contain a person. The numbers below are from a fresh evaluation of best.pt on the COCO validation set.

Setup

The model started from pretrained YOLOv8s weights and was trained for 50 epochs at 640 pixels, batch size 32, on one NVIDIA Tesla V100 (32 GB) at UCF Newton. The run took 10 hours 52 minutes. Mosaic augmentation was on, the learning rate followed a cosine schedule, and automatic mixed precision was off because it hung on these V100s. The held-out set is COCO val2017: 2,693 images and 10,777 people.

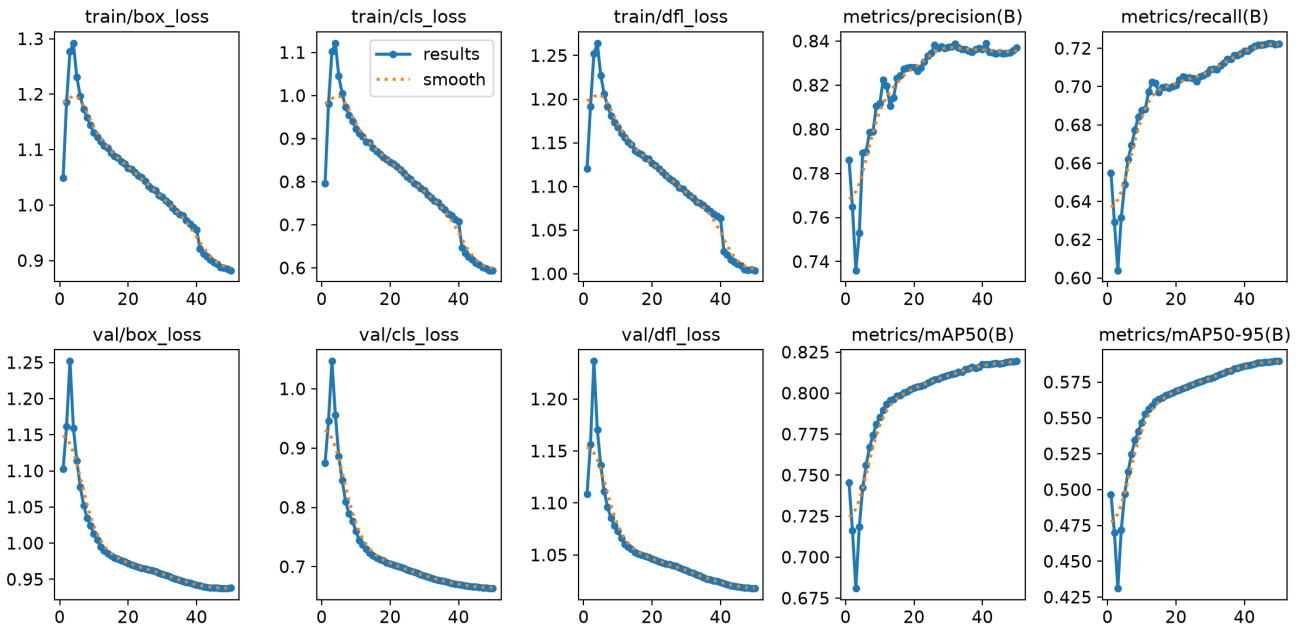
Evaluation

Metric	Result	What it means
Precision	83.50%	Of the boxes the model draws, 83.50% are people.
Recall	72.21%	The model finds 72.21% of the people in the images.
mAP@0.5	81.94%	Detection score when a box must overlap the person by half.
mAP@0.5:0.95	58.96%	Stricter score averaged from 50% overlap up to 95%.

Recall is the direct answer to how many people the model detects: 72.21% of the people in the validation images. Precision says the boxes are usually right: 83.50%. mAP@0.5 is 81.94%. The stricter mAP@0.5:0.95 is 58.96%, which drops when a box is around the person but not tight.

Training curves

Loss falls after the short warmup, and both mAP scores rise through epoch 50. Epoch 50 is the best checkpoint.



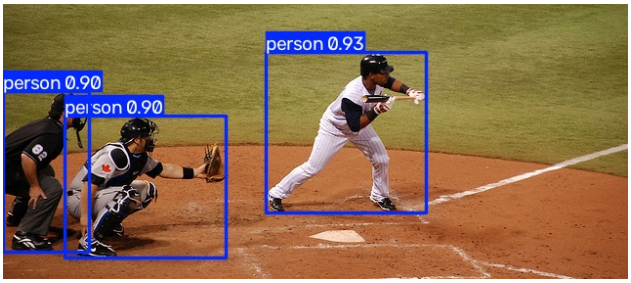
Training and validation loss, precision, recall, and mAP over 50 epochs.

Sample detections

These are validation images the model had not trained on. Boxes are the model's predictions at a confidence threshold of 0.25. The number on each box is the confidence.



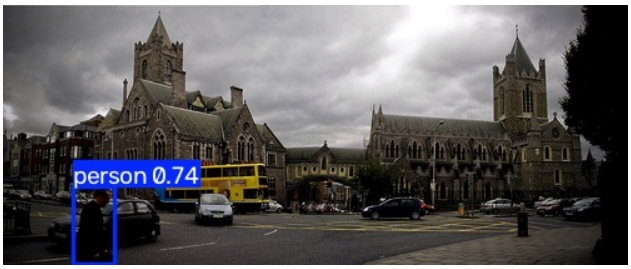
Surfer, confidence 0.79. One clear person against the water.



Baseball. Batter, catcher, and umpire, each about 0.90.



Officer on a motorcycle, confidence 0.88. The driver in the car is not boxed.



A small person at the edge of a wide street scene, confidence 0.74.

A miss and a false box

On the beach scene the people standing at the right are detected, including a partly visible person. A low-confidence box (0.42) is drawn on the flags. Small and distant people are the usual misses.



People on the right are found. One 0.42 box sits on the flags, not a person.

Weights

The checkpoint used for the numbers and the pictures is runs/yolov8s_coco_person/weights/best.pt. The final epoch is saved as last.pt in the same folder. The epoch-by-epoch table is runs/yolov8s_coco_person/results.csv.